

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 7

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- **Arrival / Departure or Input / Output curves**
- **Shockwaves**
- **Shockwave velocity equation**
- **Types of bottlenecks**
 - Temporary bottlenecks
 - Stationary bottlenecks
 - Moving bottlenecks

Modelling approaches

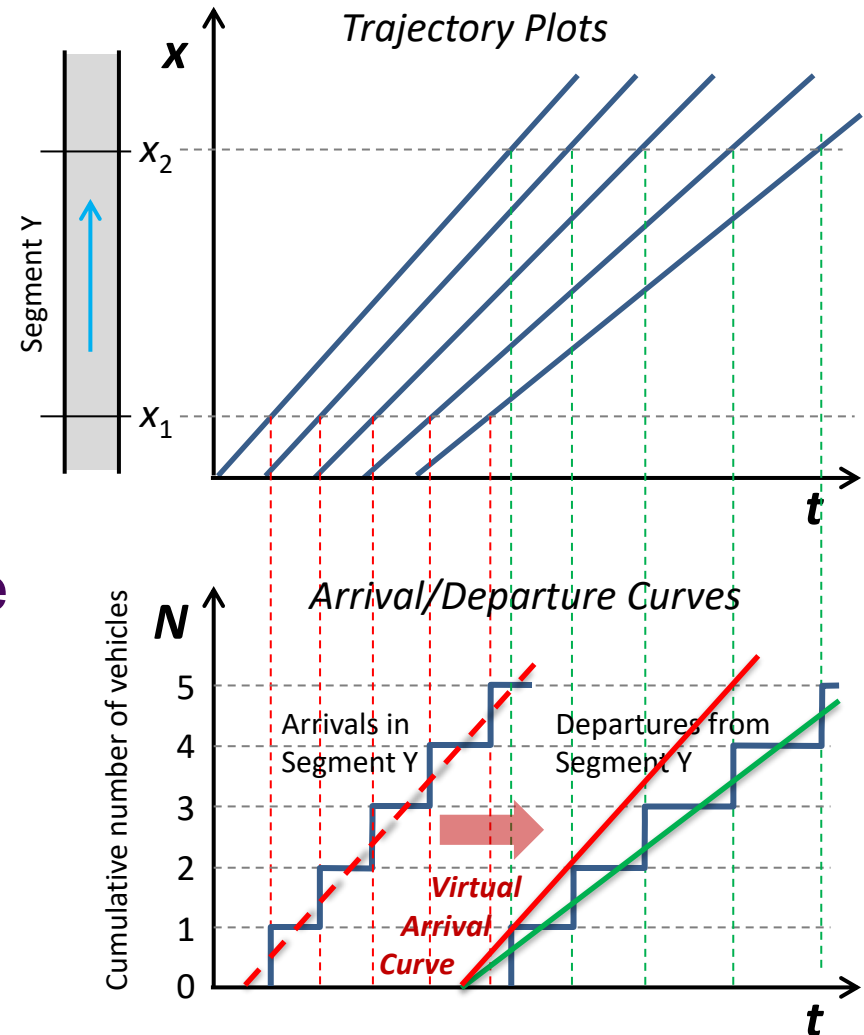
<i>Speed variables</i> <i>user variables</i>	Aggregate $u(x,t)$	Disaggregate $u_i(t)$
Aggregate $q(x,t), k(x,t)$		
Disaggregate $x_i(t)$	MESO scopic	MICRO scopic

ARRIVAL-DEPARTURE CURVES

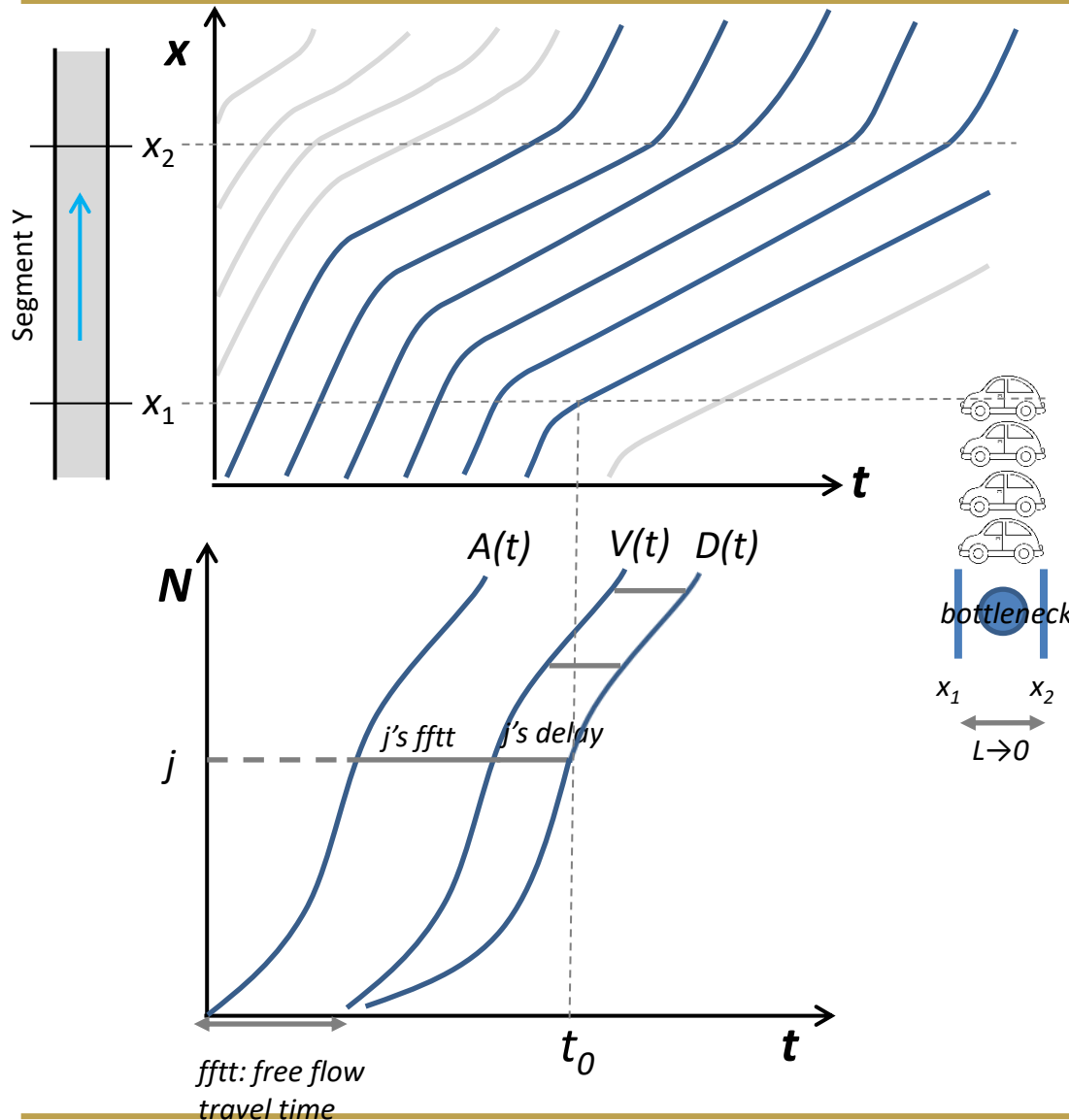
INPUT-OUTPUT CURVES

Arrival and Departure Curves

- A cumulative vehicle count curve (N -curve) shows the cumulative number of vehicles that pass a certain location x by time t .
- No entering/leaving traffic
- First-in-First-Out (FIFO)
- Introduce a virtual arrival curve by shifting the arrival curve to the right by the free-flow travel time to analyse queues and delays.



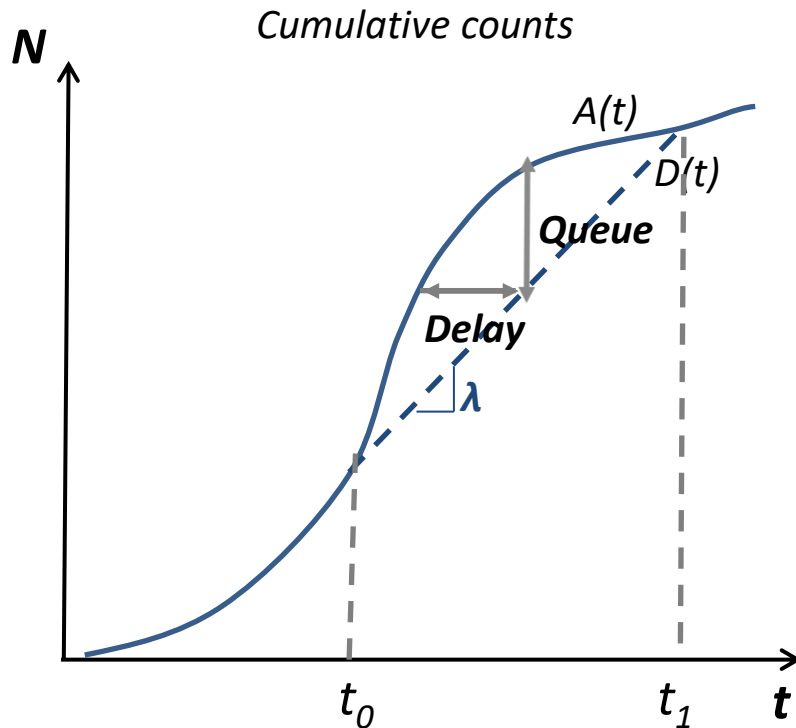
A, V and D curves



- When the N -curve is drawn on larger scale, the step function will look like a smooth function.
- If $L \rightarrow \text{large}$, $A(t) \neq V(t)$
- If $L \rightarrow 0$, $A(t) \approx V(t)$
 - $A(t) \neq D(t)$, point queue
 - $V(t)$: number of vehicles that would arrive if there was no congestion

Construct A-D curves (L~small)

Single Bottleneck



- At $t < t_0$

$$A(t) = D(t)$$
$$\dot{D}(t) = \dot{A}(t) < \lambda$$

- At $t_0 < t < t_1$

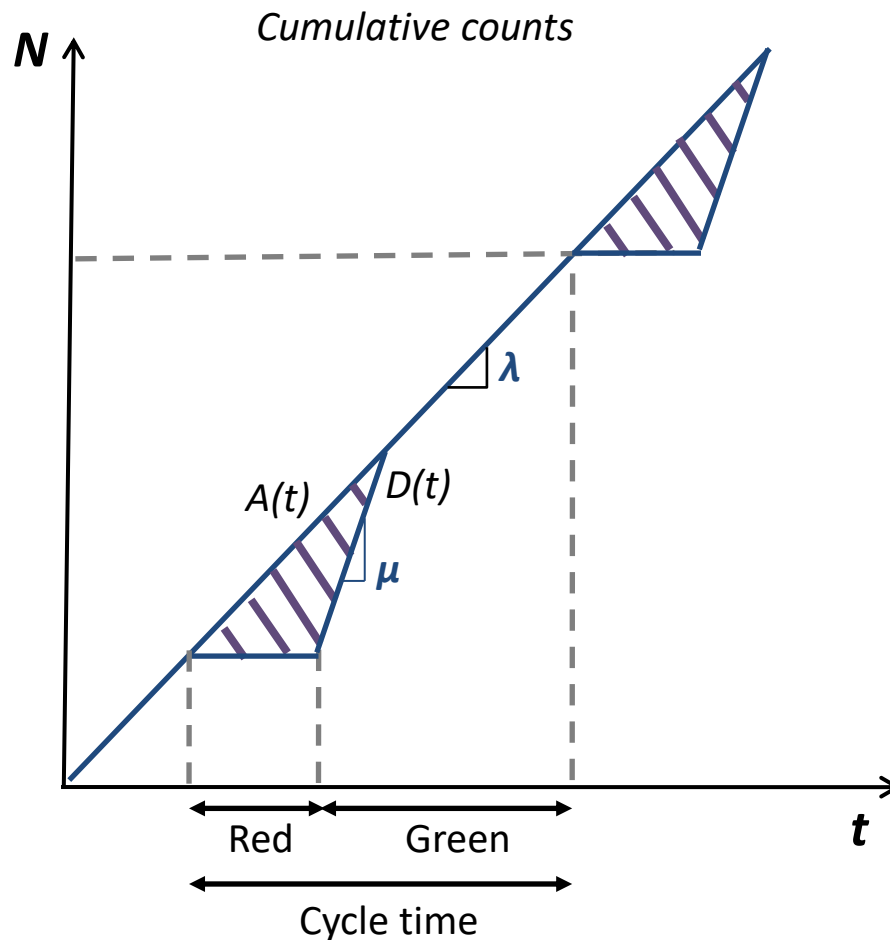
$$A(t) > D(t)$$
$$\dot{D}(t) = \lambda \text{ (service rate)}$$

Delay and Queue

- At $t > t_1$

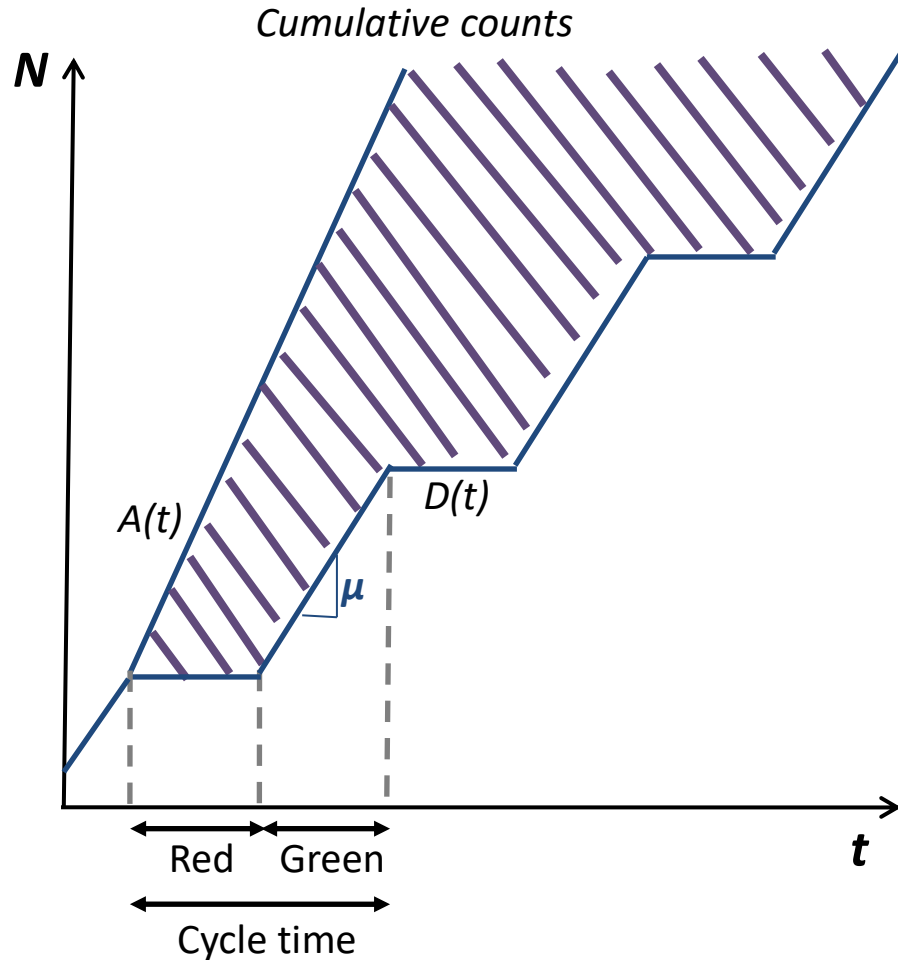
$$A(t) = D(t)$$
$$\dot{D}(t) = \dot{A}(t)$$

A-D (Traffic signal)



- Delay because of a traffic signal
- Traffic flows at rate λ
- Saturation flow is μ
- $\lambda \ll \mu$
- What if queue cannot be emptied?

A-D (Traffic signal)



- Delay because of a traffic signal
- Traffic flows at rate λ
- Saturation flow is μ
- $\lambda \gg \mu$
- What if queue cannot be emptied?
- **SPILLBACK!!**

Question

Queue clearance

Traffic flows at rate λ , while saturation flow is μ .

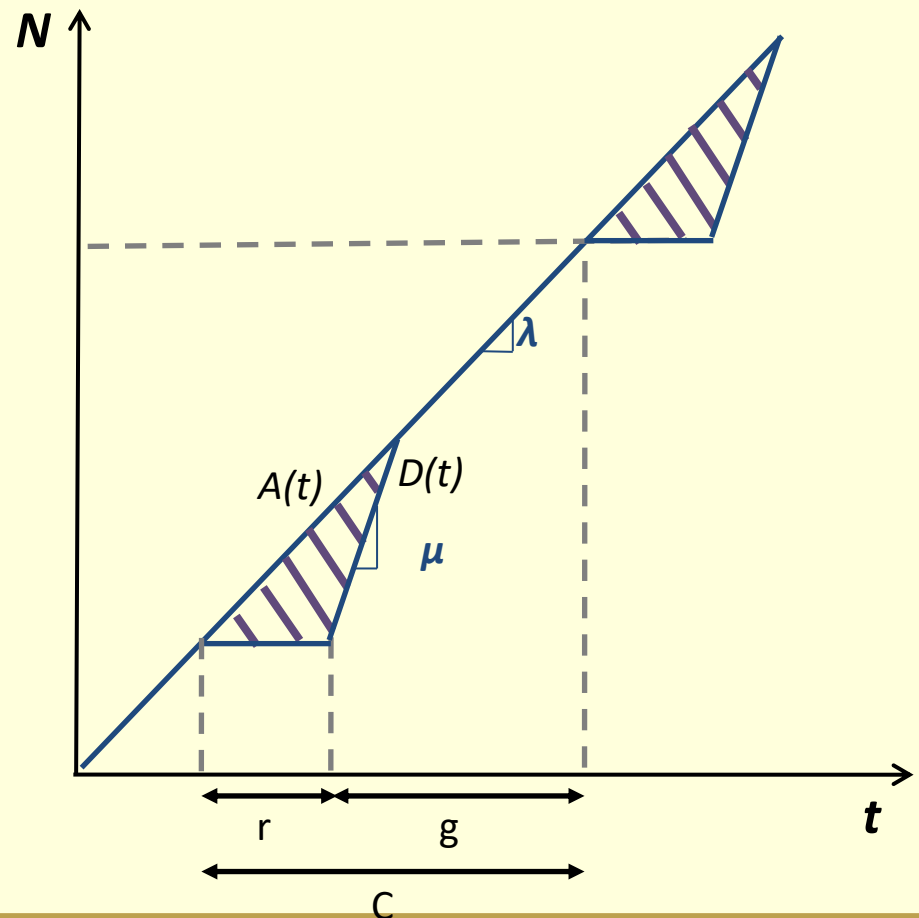
Assume red phase length is r , green phase length is g , and cycle time is C , which one of the following conditions is necessary for the queue to be emptied at the end of the cycle?

A. $\lambda C \leq \mu r$

B. $\lambda C \leq \mu g$

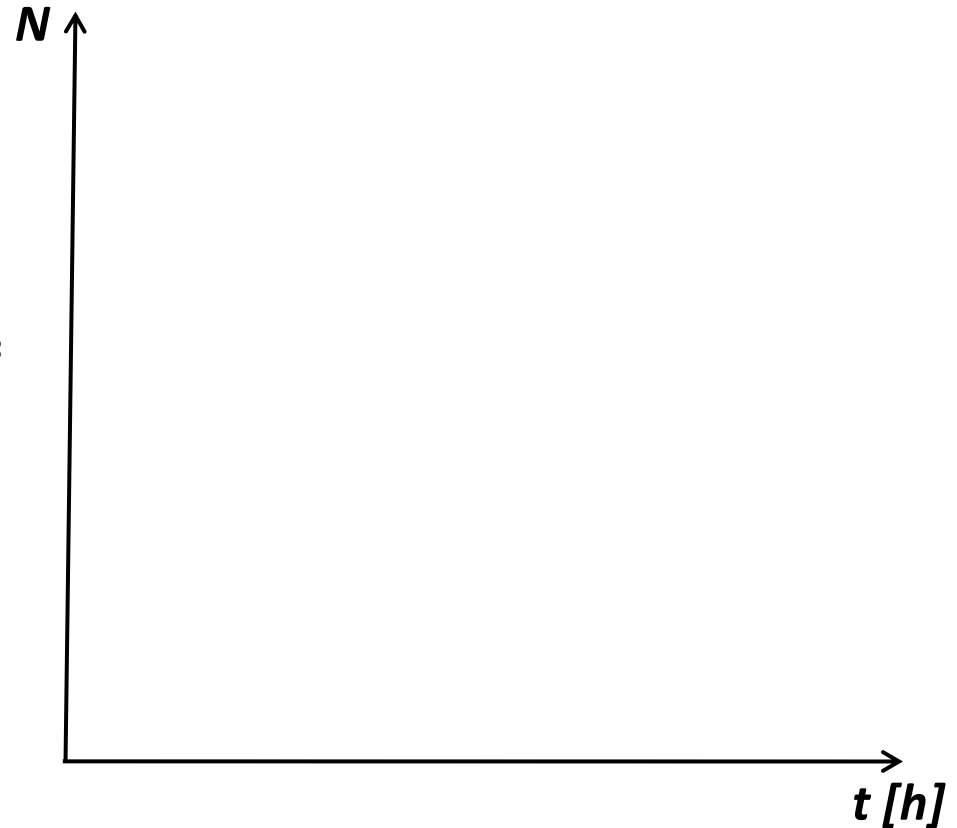
C. $\lambda r \geq \mu C$

D. $\lambda g \geq \mu g$



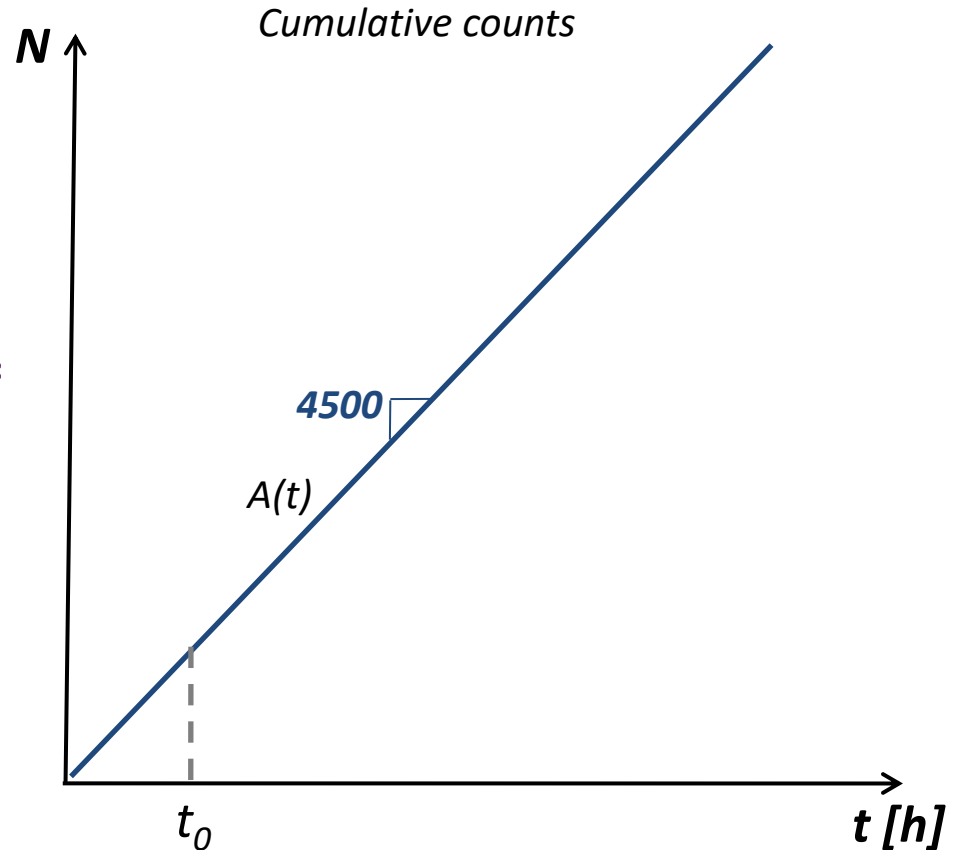
Exercise (A-D)

- A freeway with 3 lanes and maximum capacity $C = 6000$ veh/h
- Demand = $\frac{3}{4} * C$
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance



Exercise (A-D)

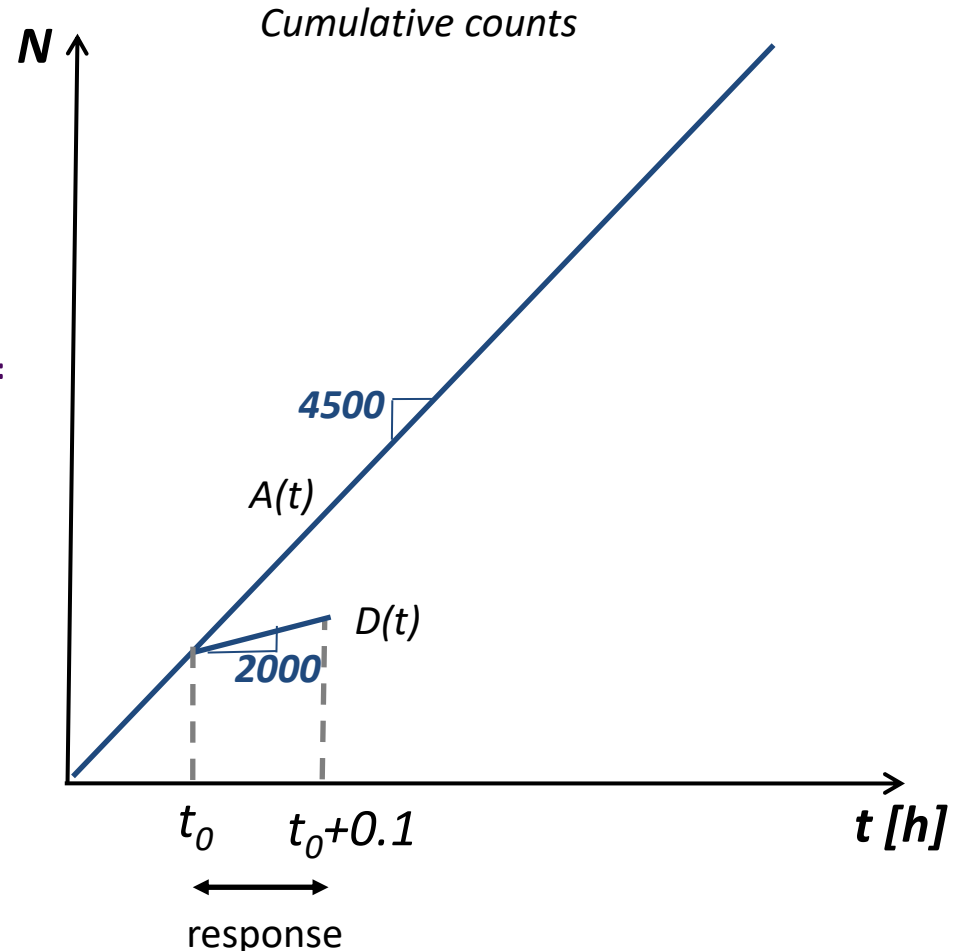
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- Two lanes remain open during clearance
- Between t_0 and $t_0+0.1$

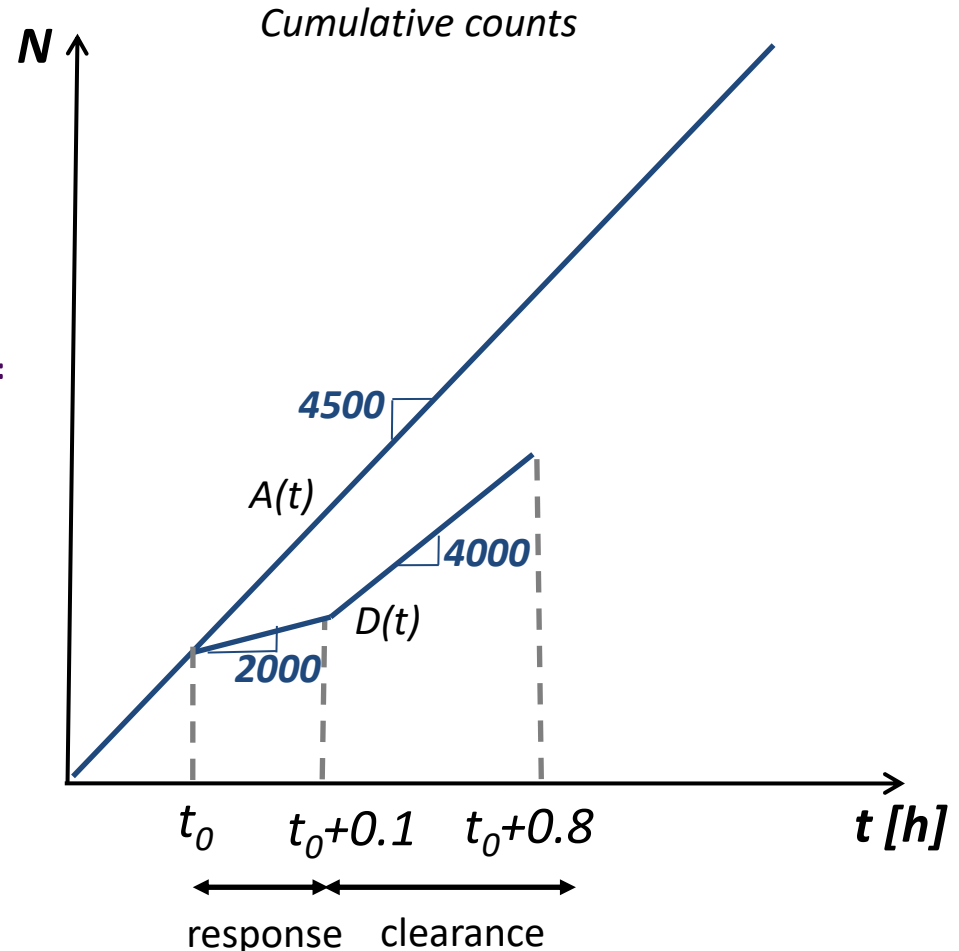
$$\dot{D}(t) = \frac{C}{3} = 2000 \text{ veh/h}$$



Exercise (A-D)

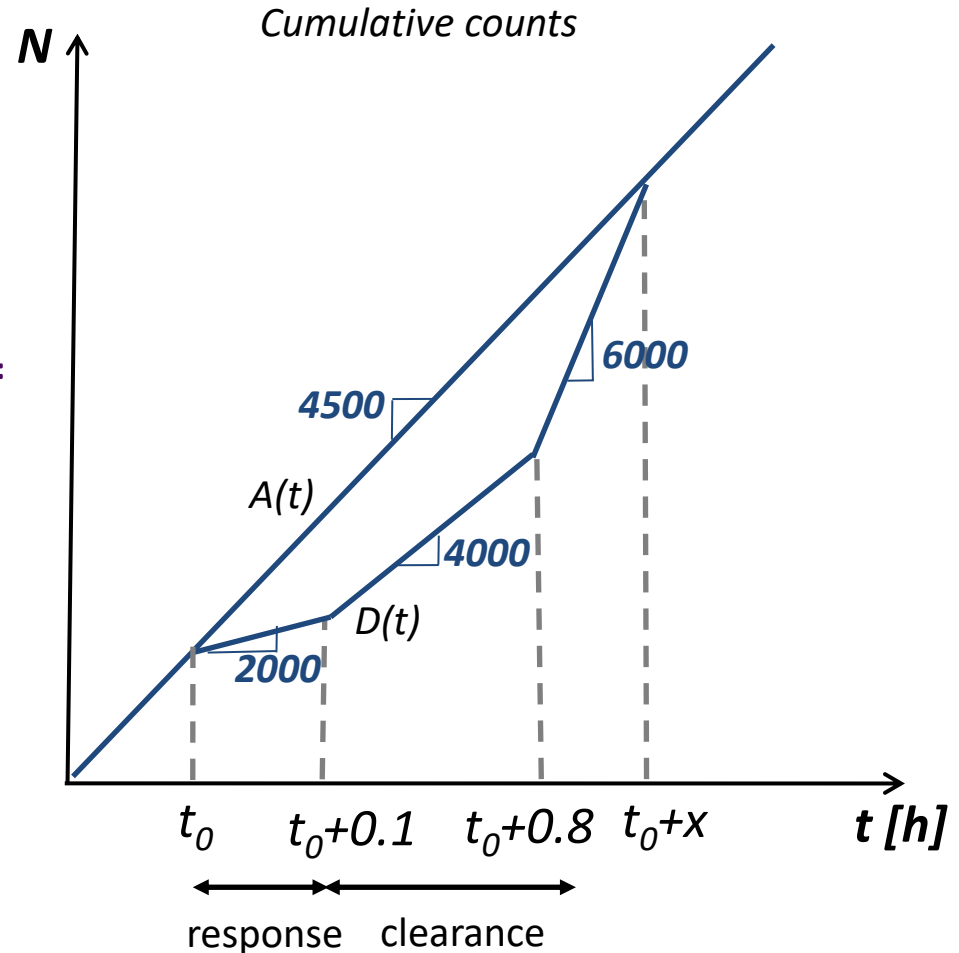
- A freeway with 3 lanes and maximum capacity $C = 6000$ veh/h
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- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance
- Between $t_0 + 0.1$ and $t_0 + 0.8$

$$\dot{D}(t) = \frac{2C}{3} = 4000 \text{ veh/h}$$



Exercise (A-D)

- A freeway with 3 lanes and maximum capacity $C = 6000$ veh/h
- Demand = $\frac{3}{4} * C$
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance



Exercise (A-D)

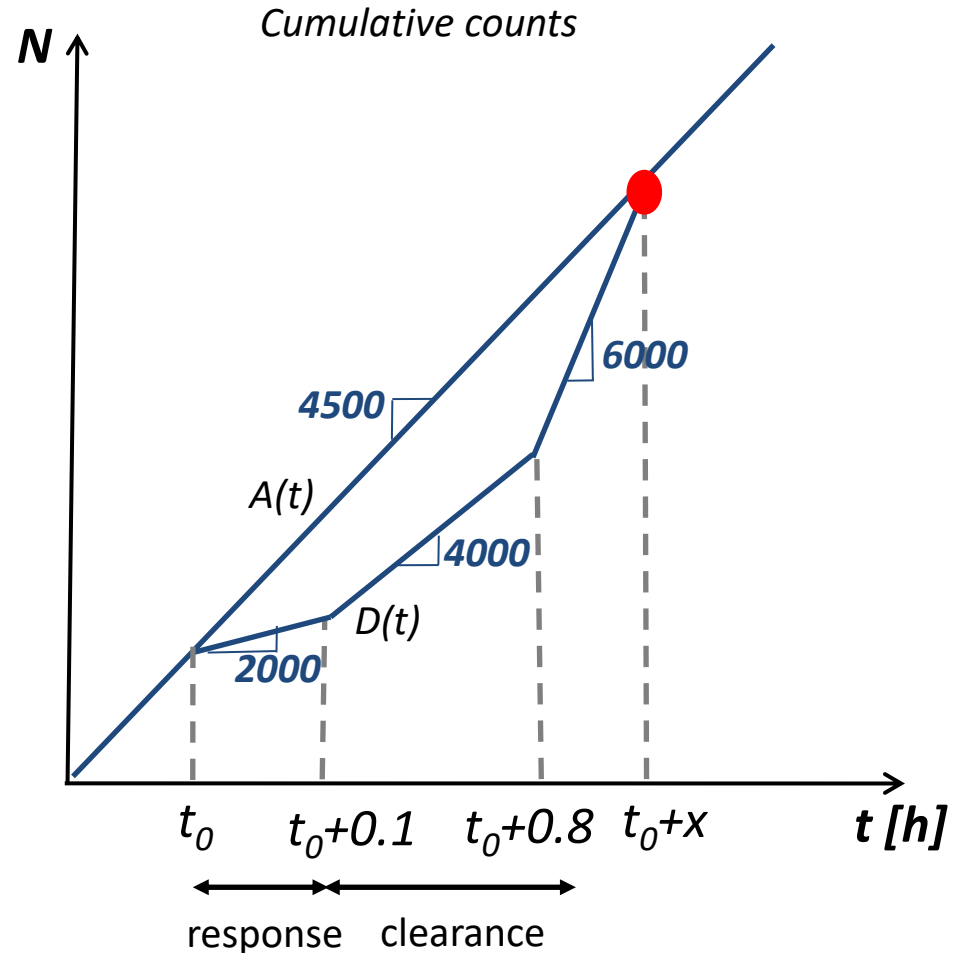
Duration of congestion

$x = ?$

$4500 * x =$

$2000 * 0.1 + 4000 * 0.7 + 6000 * (x - 0.8)$

$x = 1.2 \text{ h}$



Exercise (A-D)

$$\text{Total delay} = w_{\text{tot}} = A1 + A2 + A3$$

$$n1 = (4500 \cdot 0.1 - 2000 \cdot 0.1) = 250 \text{ veh}$$

$$n2 = (4500 \cdot 0.8 - 2000 \cdot 0.1 - 4000 \cdot 0.7)$$

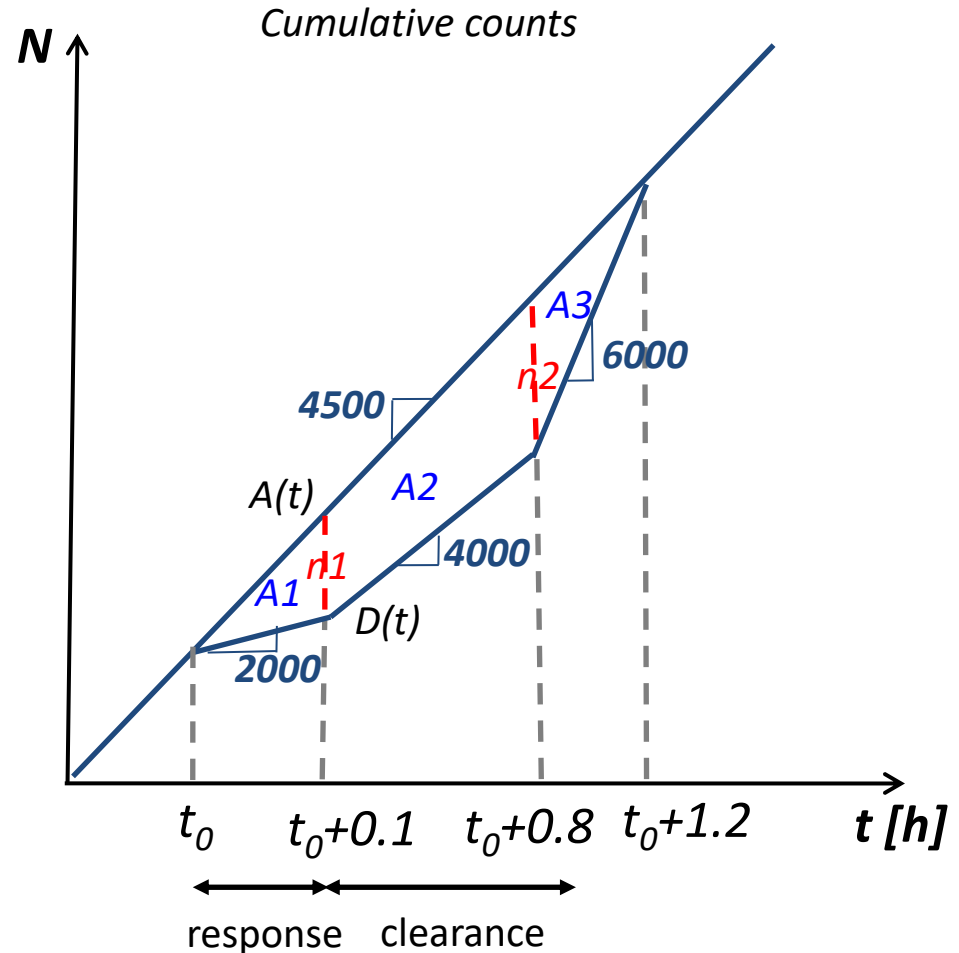
$$= 600 \text{ veh}$$

$$A1 = n1 \cdot 0.1 / 2 = 12.5 \text{ veh.h}$$

$$A2 = (n1 + n2) / 2 \cdot 0.7 = 297.5 \text{ veh.h}$$

$$A3 = n2 \cdot 0.4 / 2 = 120 \text{ veh.h}$$

$$w_{\text{tot}} = 430 \text{ veh.h}$$



Exercise (A-D)

Maximum delay = $w_{\max}$?

Maximum queue = $N_{\max}$?

$N_{\max} = n2 = 600$ veh

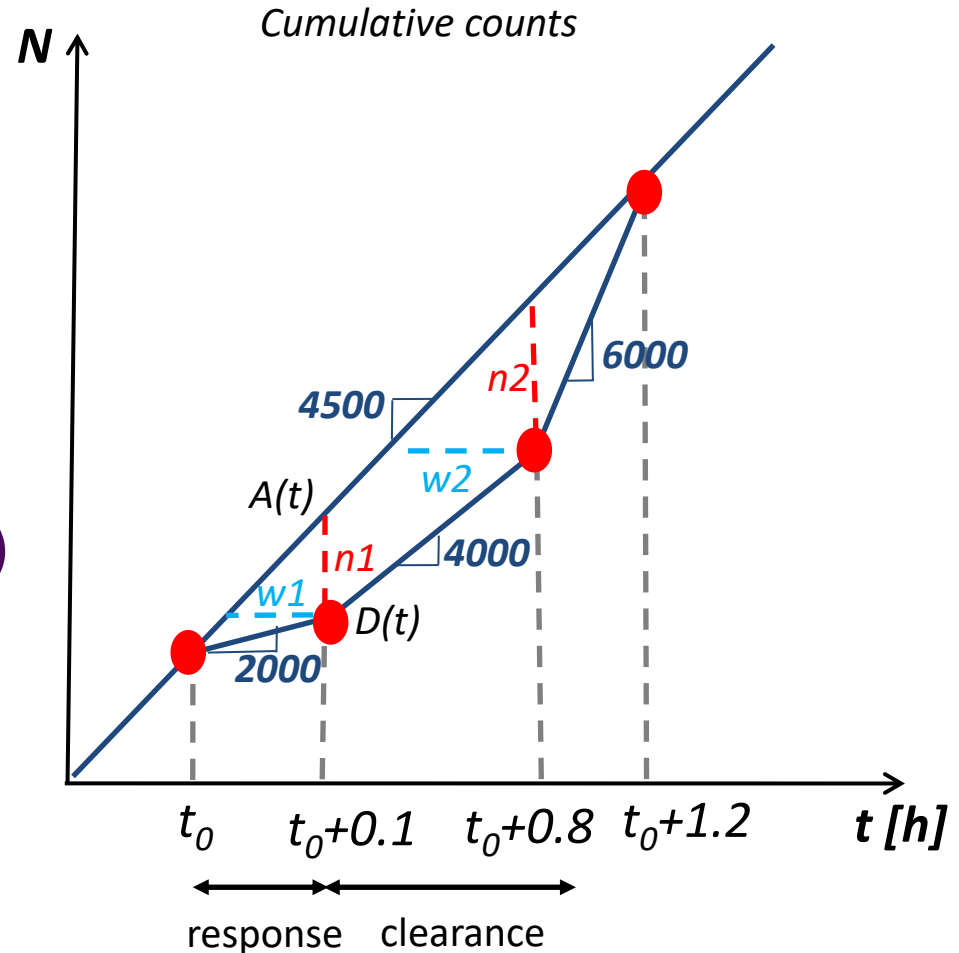
$2000 * 0.1 = 4500(0.1 - w1)$

$w1 = 0.055$ h

$2000 * 0.1 + 4000 * 0.7 = 4500(0.8 - w2)$

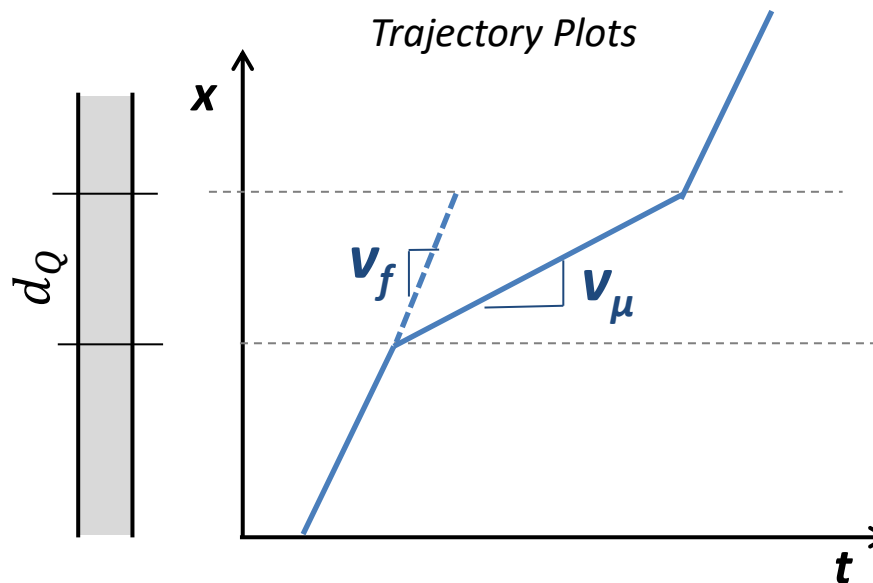
$w2 = 0.133$ h

$w_{\max} = w2 = 0.133$ h



End of queue

- Distance travelled while in the queue



$$w = \frac{d_Q}{v_\mu} - \frac{d_Q}{v_f}$$

$$d_Q = \frac{w}{\frac{1}{v_\mu} - \frac{1}{v_f}}$$

d_Q = distance of queue [km]

v_f = free flow speed [km/h]

v_μ = actual speed [km/h]

w = delay [h]

End of queue

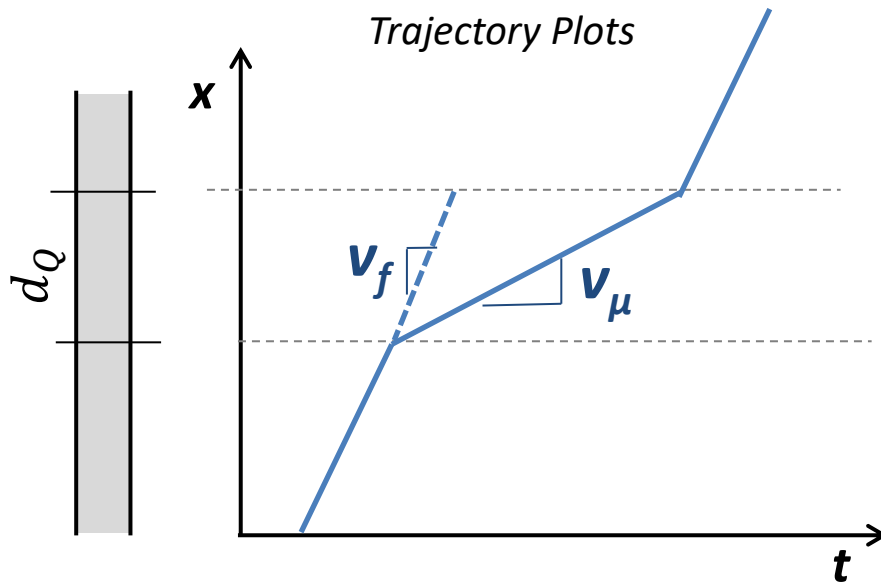
- Time spent in the queue

$$t_Q = \frac{d_Q}{v_\mu} = \frac{d_Q}{v_f} + w$$

$$t_Q = \frac{w}{\frac{1}{v_\mu} - \frac{1}{v_f}} + w$$

$$t_Q = \frac{w \cdot v_\mu}{v_f - v_\mu} + w$$

$$t_Q = \frac{w}{1 - \frac{v_\mu}{v_f}}$$



Question 2

Input-Output

Calculate maximum length of the queue and maximum time spent in the queue based on the below A-D curve?

$n_1=250$ veh, $n_2=600$ veh

$w_1 = 0.055$ h, $w_2=0.133$ h

$v_\mu=28.57$ km/h, $v_f=100$ km/h

A. 5.00km, 0.25 h

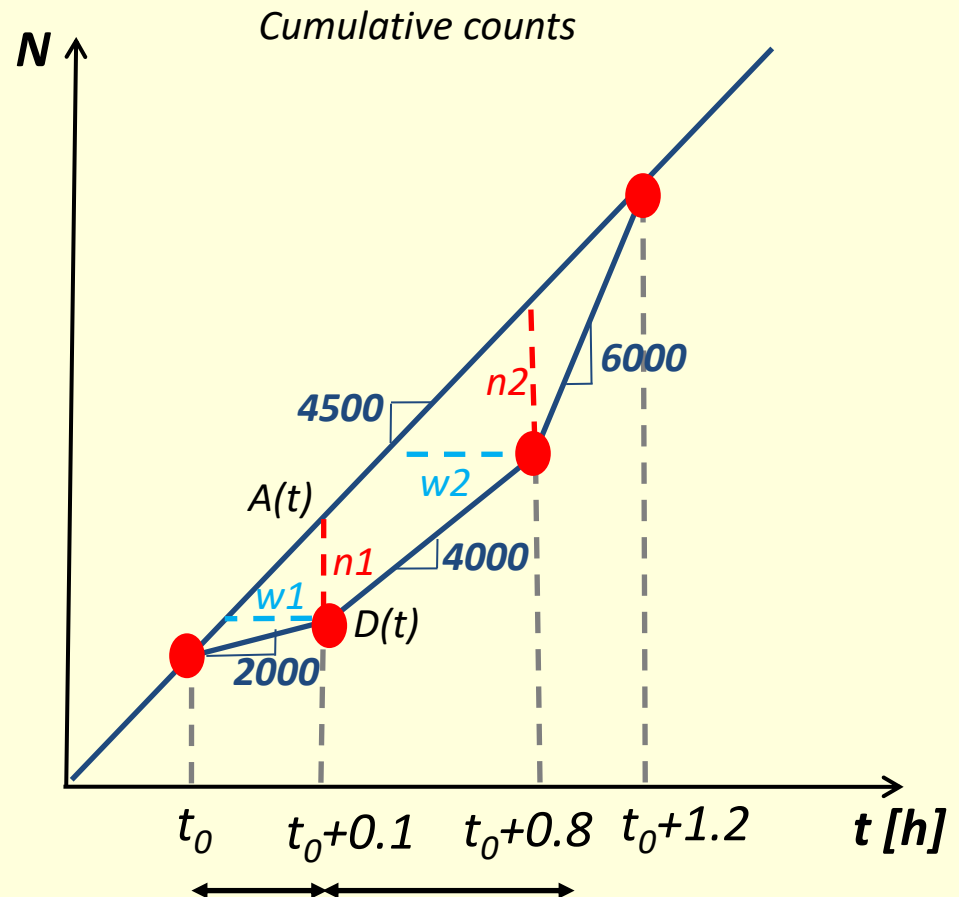
B. 4.25km, 0.10 h

C. 3.80km, 0.35 h

D. 5.32km, 0.19 h

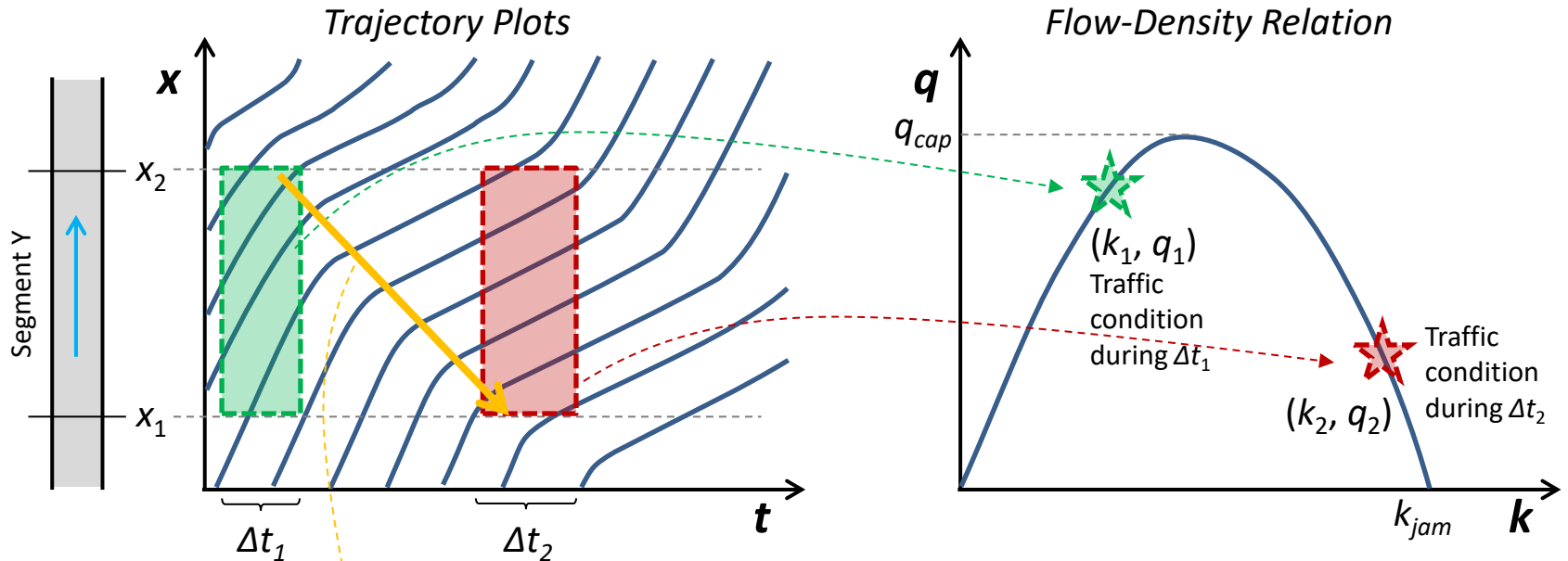
E. 4.70km, 0.35 h

$$d_Q = \frac{w}{\frac{1}{v_\mu} - \frac{1}{v_f}} \quad t_Q = \frac{w}{1 - \frac{v_\mu}{v_f}}$$



SHOCKWAVES

Space-Time Diagram (x-t plot) vs. Fundamental Diagram (q-k plot)



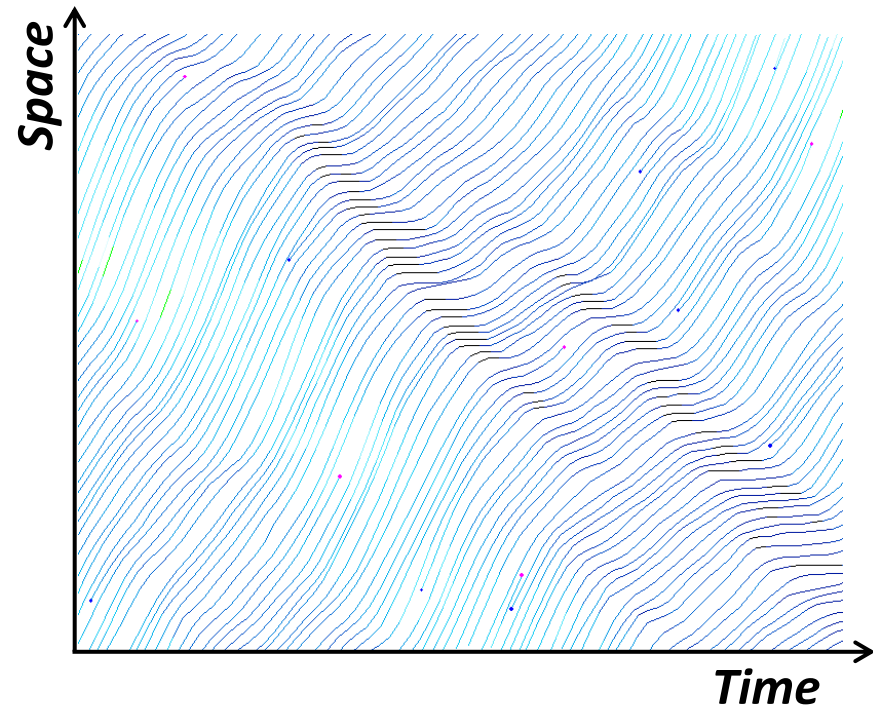
Points on the fundamental diagram (FD) describe possible traffic conditions on the road segment Y.

→ *FD is a (steady-state) property of the road.*

Transition between state 1 and 2?

→ **Shockwave**

- “**Shockwave**” – a motion or propagation of change in traffic conditions (flow, density, and speed)
 - Whenever a stream of traffic under certain flow conditions (q_1 , k_1 and v_1) meets another stream under different conditions (q_2 , k_2 and v_2), a **shock wave** is formed.
- A moving boundary between two different traffic states
 - a boundary in the space-time domain that shows a **discontinuity** in traffic stream conditions.
- The shock front may **move forward or backward or stay at the same place** with respect to the road.



■ Phantom traffic jams or stop-and-go waves

- <https://www.youtube.com/watch?v=goVjVVaLe10>

■ How we can use this information

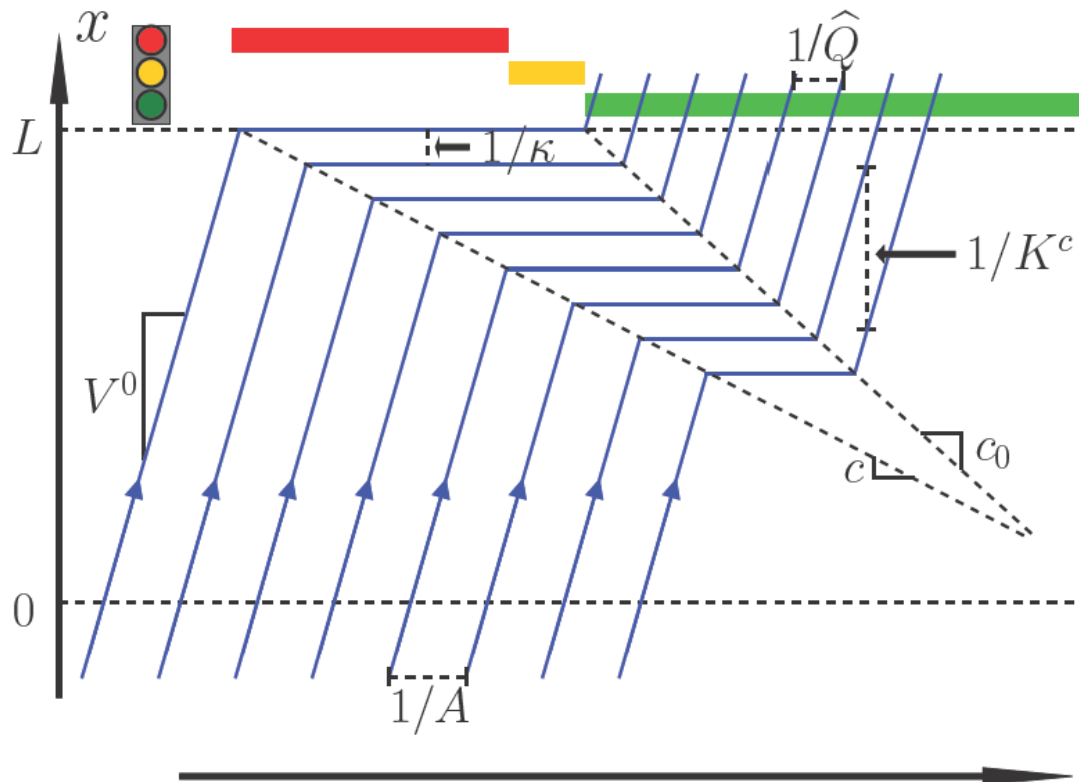
- <https://www.youtube.com/watch?v=rdJFeUPc2gg>

■ Simulation tools

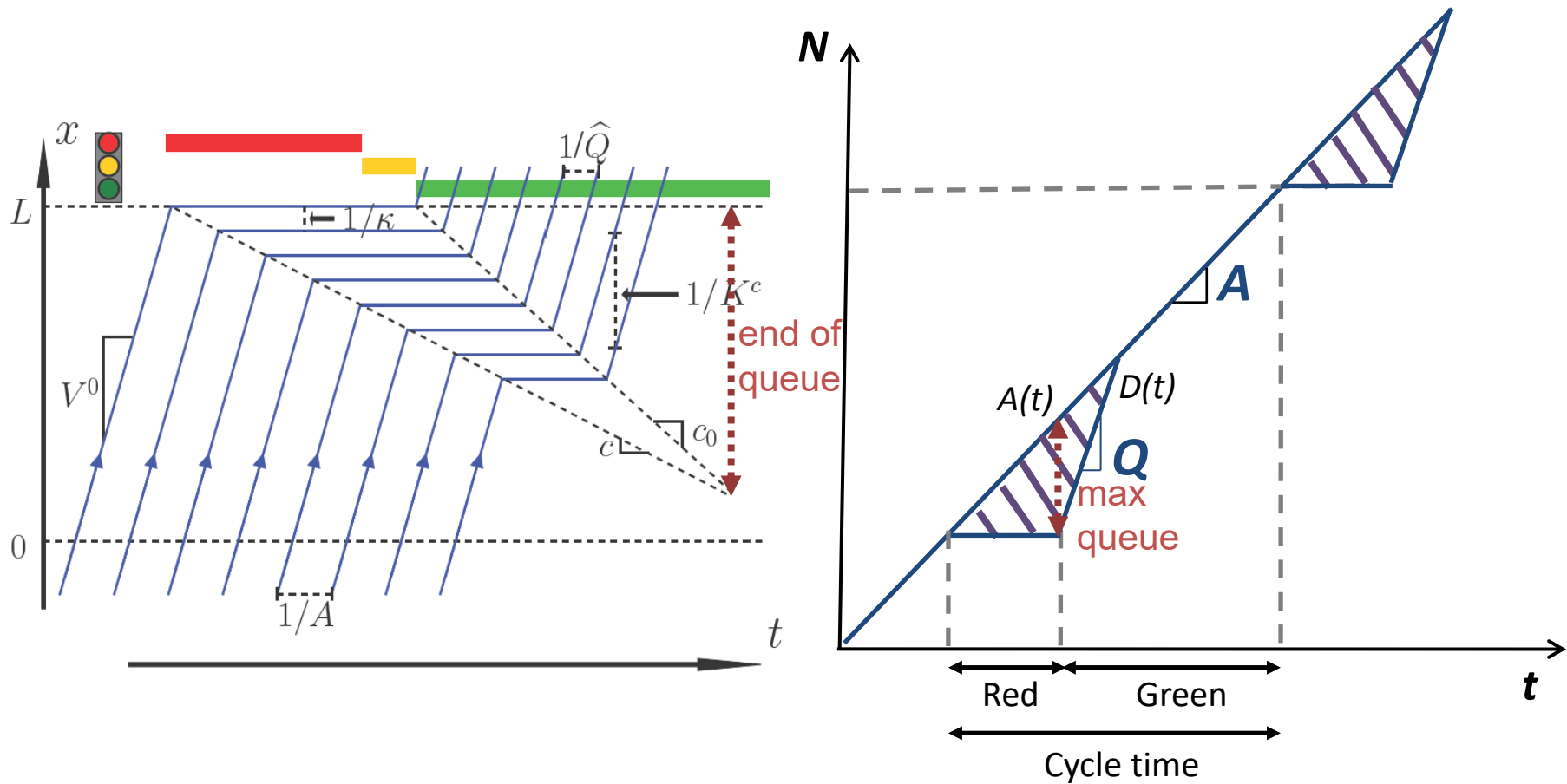
- <https://setosa.io/blog/2014/09/02/gridlock/>
- <http://www.traffic-simulation.de/index.html>

Traffic signal

- “*Shockwave*” – caused by red and green phases
 - Immediate change of speeds along the shockwaves
 - Infinite acceleration – limitation of modelling approach



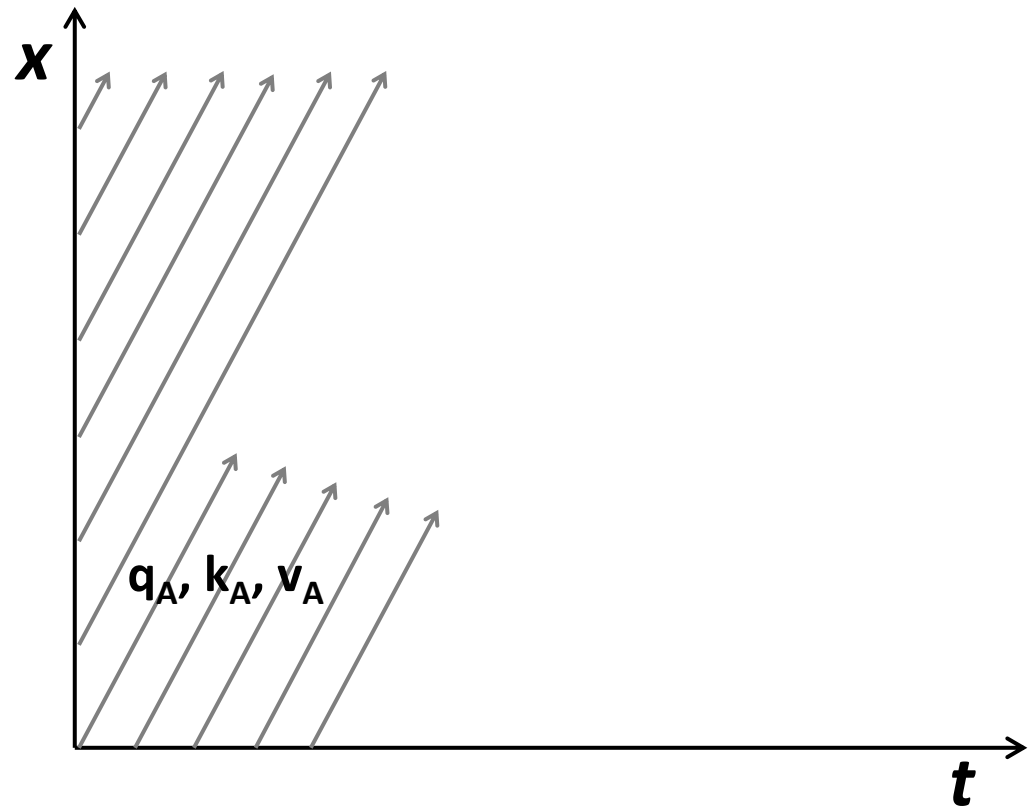
Traffic signal



Shockwave Example

■ Formation and Propagation Scenarios [1/6]

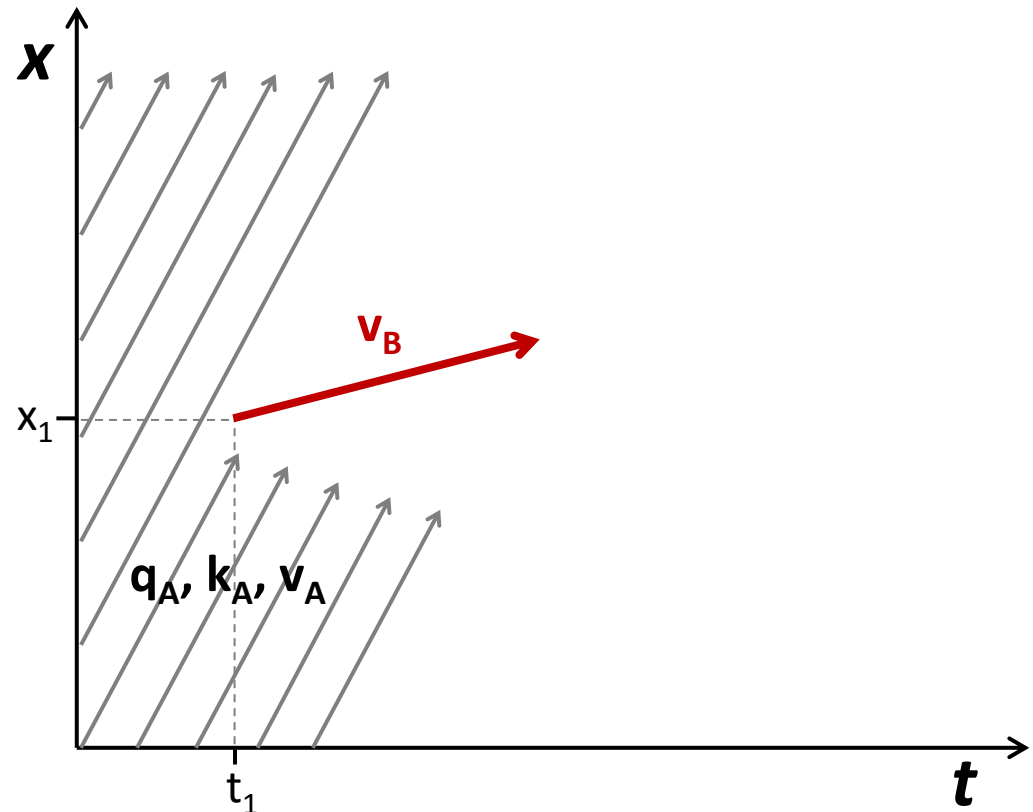
- Traffic stream moving at flow q_A , density k_A , and speed v_A .



Shockwave Example

■ Formation and Propagation Scenarios [2/6]

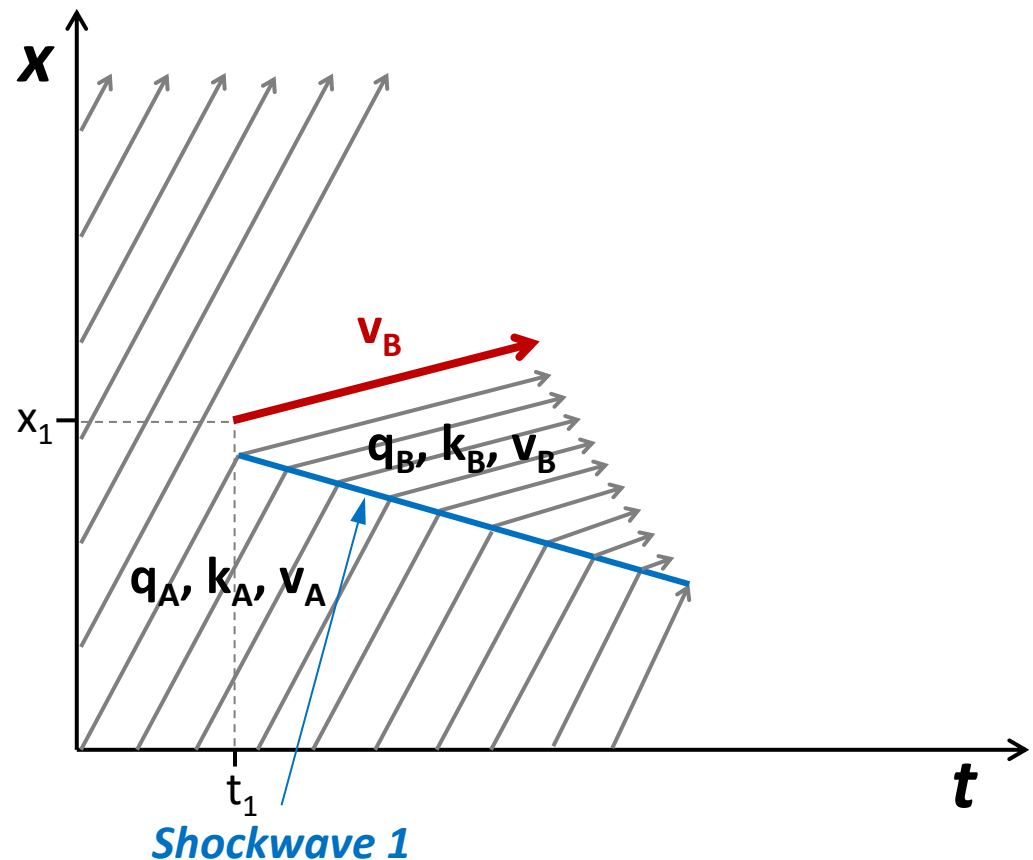
- Traffic stream moving at flow q_A , density k_A , and speed v_A .
- At time t_1 and point x_1 , a slow moving vehicle (red trajectory) with speed v_B enters a traffic stream.



Shockwave Example

■ Formation and Propagation Scenarios [3/6]

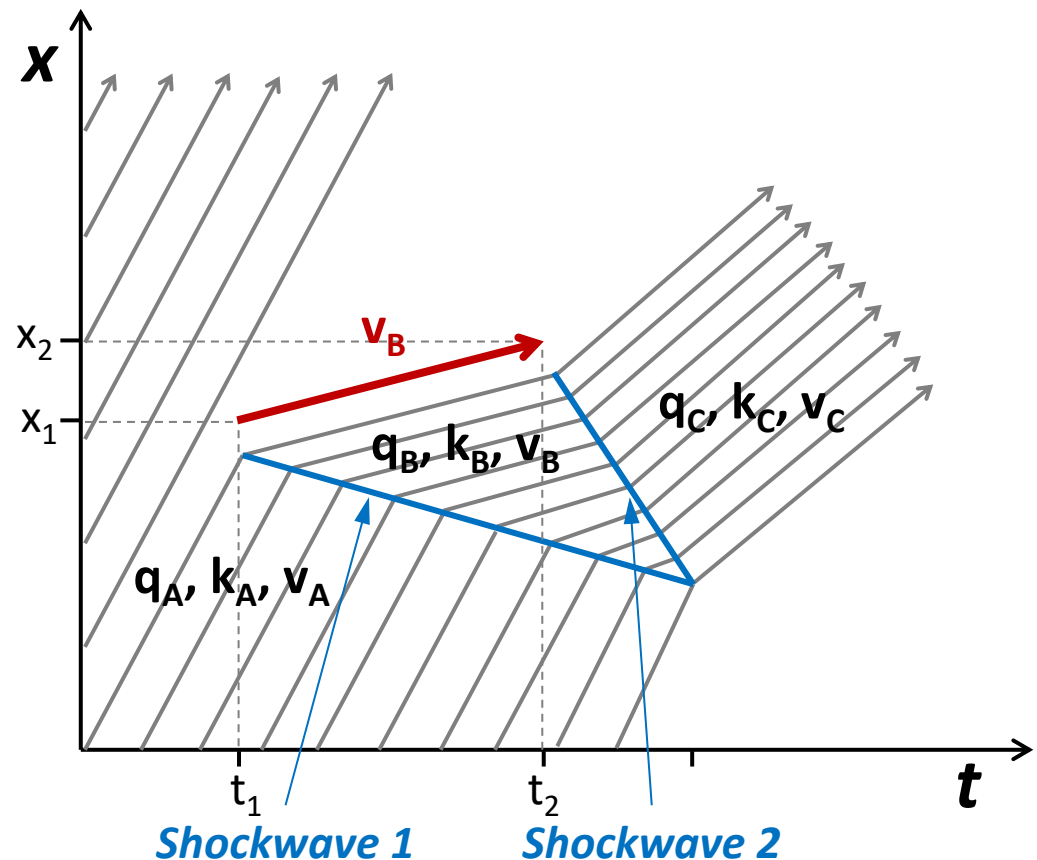
- Traffic stream moving at flow q_A , density k_A , and speed v_A .
- At time t_1 and point x_1 , a slow moving vehicle (red trajectory) with speed v_B enters a traffic stream.
- The slow moving vehicle slows the traffic and creates another stream condition q_B , k_B , and v_B .
- When the stream under A meets the stream under B, **shockwave 1** is formed and propagates upstream.



Shockwave Example

■ Formation and Propagation Scenarios [4/6]

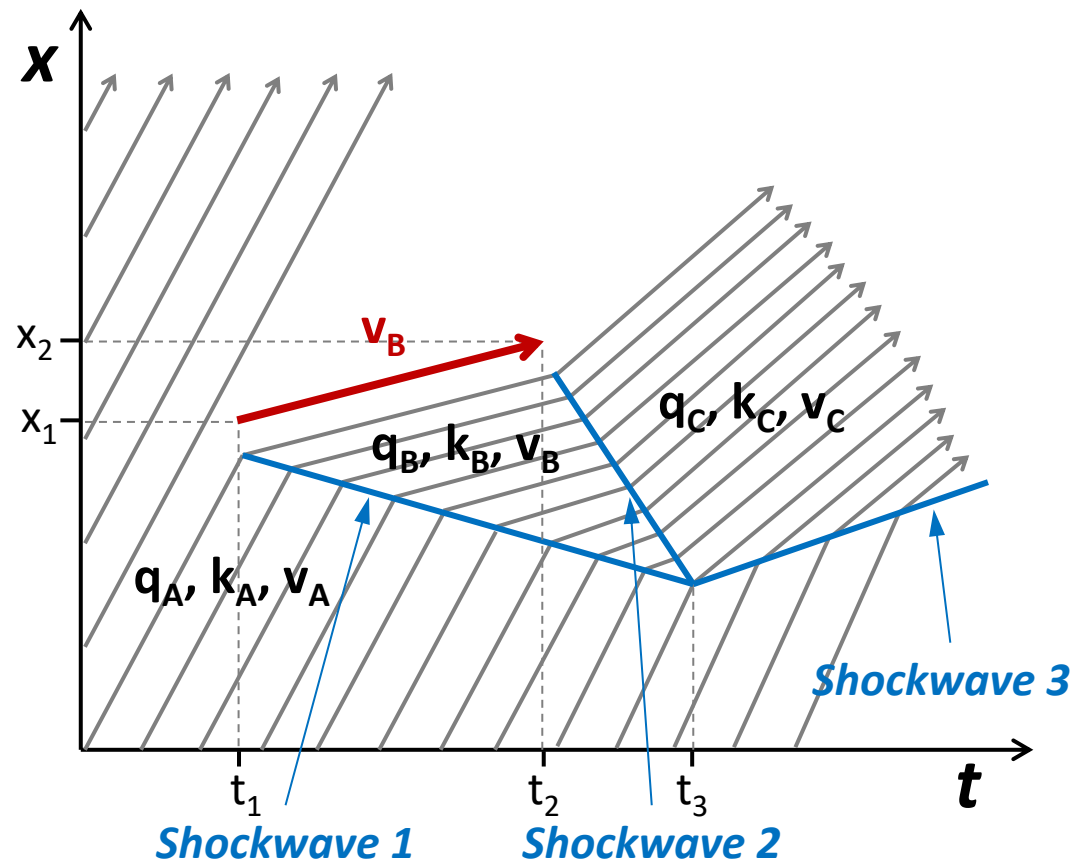
- At time t_2 and point x_2 , a **slow moving vehicle (red trajectory)** leaves the **traffic stream** and the congested conditions is released at some other stream condition q_C, k_C , and v_C .
- When the stream under B meets the stream under C, **shockwave 2** is formed and propagates upstream.



Shockwave Example

■ Formation and Propagation Scenarios [5/6]

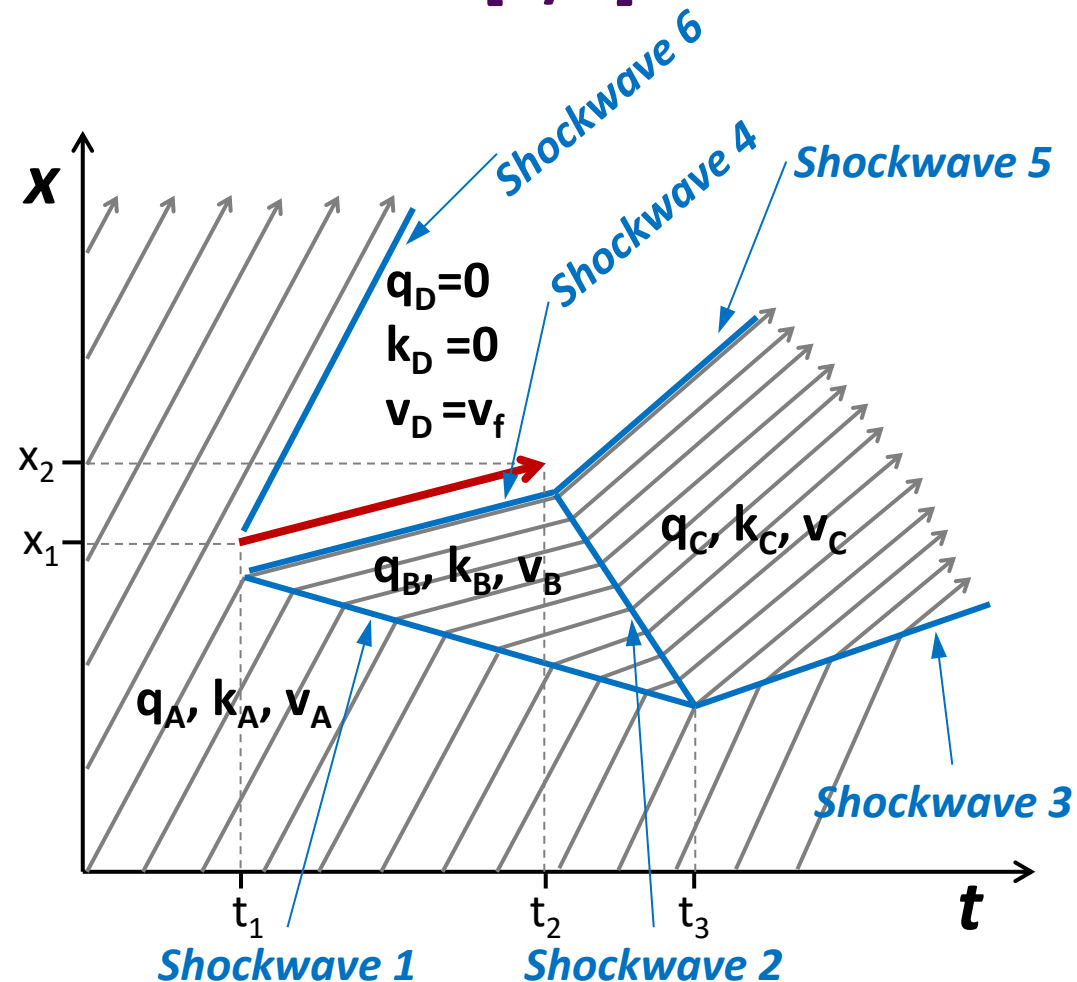
- At time t_3 , these **two shock waves (shockwaves 1 and 2) meet** and the traffic condition B ends.
- At this point, the traffic condition A meets condition C, causing **shockwave 3**.



Shockwave Example

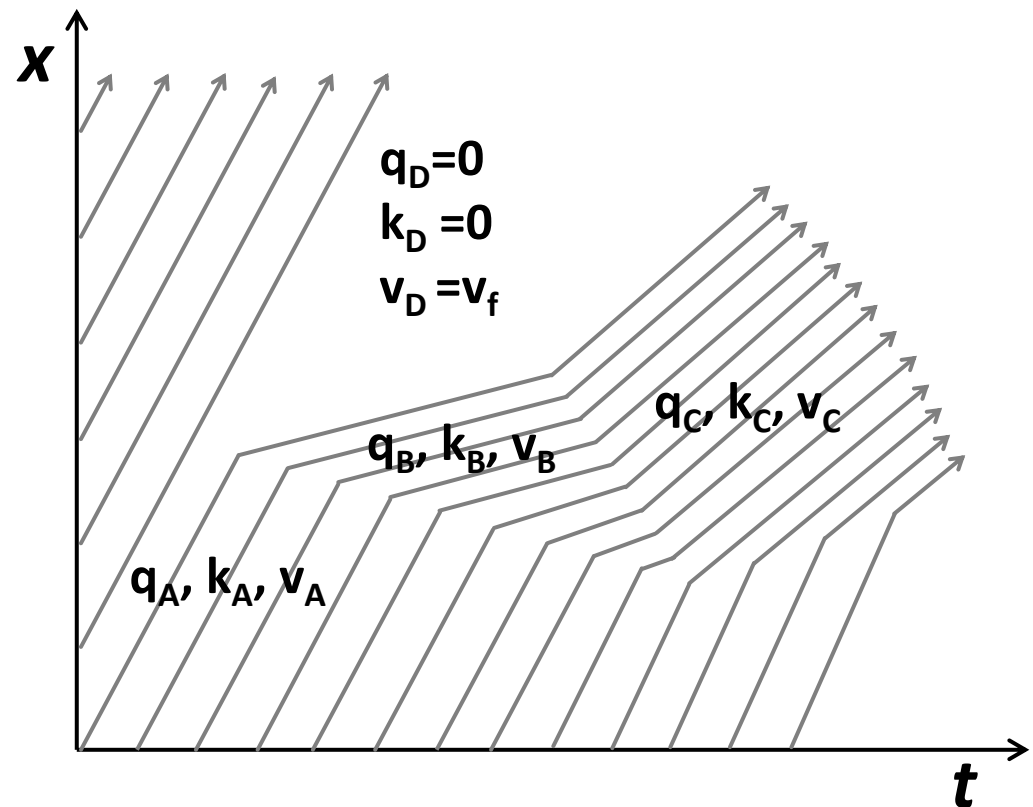
■ Formation and Propagation Scenarios [6/6]

- Area D indicates a stream with **the free-flow conditions**: $q_D=0$, $k_D=0$, and $v_D = v_f$
- **Shockwave 4** is formed at the boundary between conditions B and D and moves with the slow moving vehicle.
- **Shockwave 5** is formed at the boundary between conditions C and D and moves with the first vehicle that is released.
- **Shockwave 6** is formed at the boundary between conditions A and D and moves with the last vehicle that avoided slow moving vehicle.



■ Key Parameters of a Shock Wave

- Identify traffic states
- the point at which it starts,
- the point at which it ends, and
- **the velocity of the shock wave**
(speed + direction).



Velocity of Shockwaves

Shockwave Velocity (w)

- Expansion of the state 2 during T in [veh]:

$$(q_1 T - q_2 T - w \times T \times k_1)$$

q_1 : inflow, q_2 : outflow,

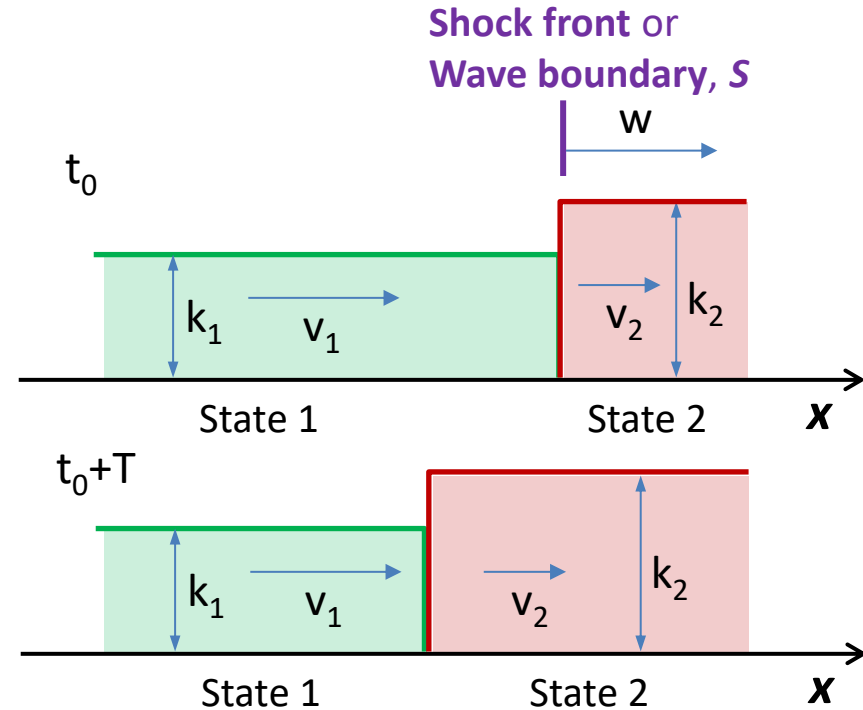
- $w \times T$: space taken over by state 2

- This is the same as;

$$- w \times T \times k_2$$

$$(q_1 - q_2 - w \times k_1) \times T = - w \times k_2 \times T$$

$$w = \frac{q_1 - q_2}{k_1 - k_2} = \frac{q_2 - q_1}{k_2 - k_1} \quad [\text{km/hr}]$$



▪ Shockwave Velocity (w)

$$w = \frac{q_2 - q_1}{k_2 - k_1}$$

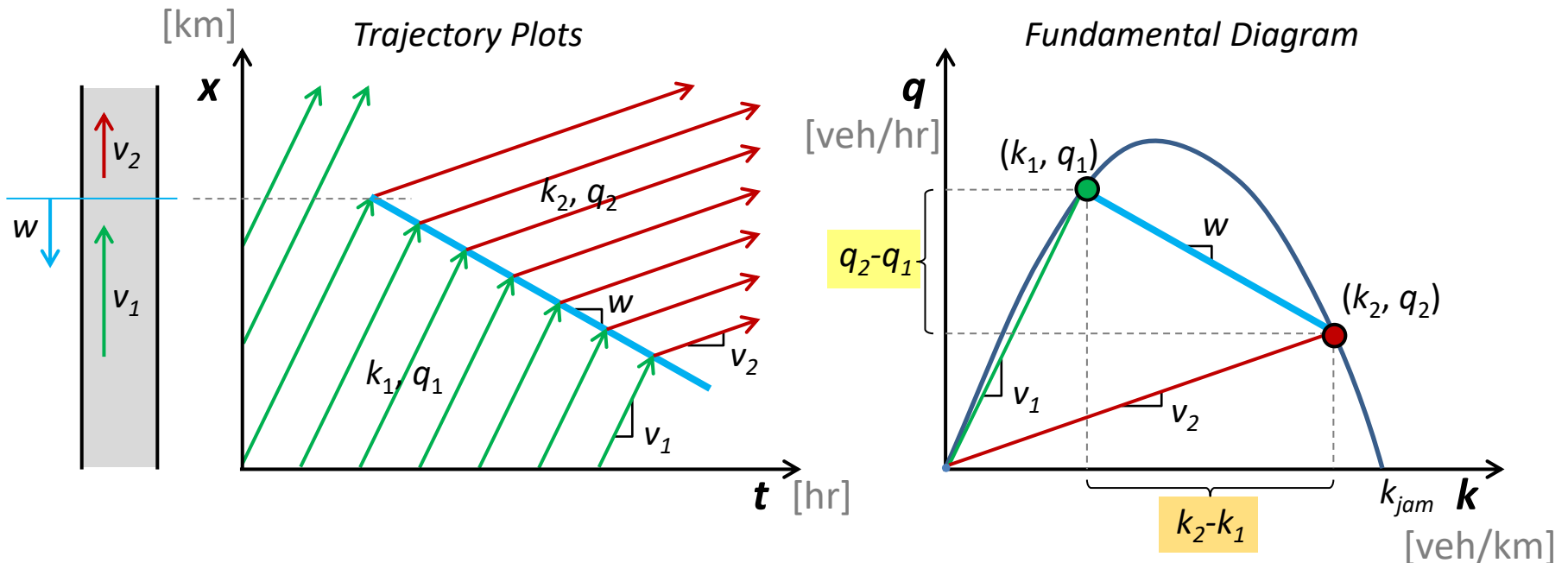
- is the ratio of the change in flow over the shock front S to the change in density over the shock front S .
- $w > 0$ (+) : Shock wave is travelling in same direction as traffic stream.
- $w < 0$ (-) : Shock wave is travelling upstream or against the traffic stream.

Velocity of Shockwaves

Shockwave Velocity (w)

$$w = \frac{q_2 - q_1}{k_2 - k_1} \quad [\text{km/hr}]$$

Geometric Interpretation on Fundamental Diagram



Shockwave Velocity

The traffic flow on a highway is $q=2100$ veh/h with speed of $v=70$ km/h. As a result of an accident, the road is blocked and vehicles arriving after the accident must stop. The density in the queue is $k=240$ veh/km (jam density). **What is the speed of shockwave, w (km/h), generated by this accident?**

- A. 10 km/h
- B. 0 km/h
- C. - 5 km/h
- D. - 8 km/h
- E. - 10 km/h

$$w = \frac{q_2 - q_1}{k_2 - k_1}$$

Queue growth rate

Traffic conditions within the impacted area are $k_2 = 240$ veh/km and $v_2 = 0$ veh/km. The capacity of the system is 2800 veh/h, free flow speed is 70 km/h, and the fundamental diagram is triangular. The accident is cleared after half an hour. **What is the speed of second shockwave due to incident clearance?**

[Solution]

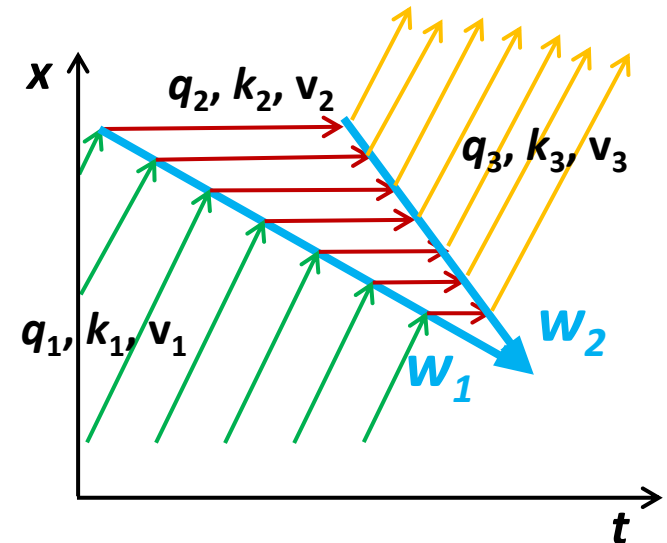
$q_3 = 2800$ veh/h, $v_3 = 70$ km/h
→ $k_3 = 2800/70 = 40$ veh/km

The shockwave speed

$$w_2 = \frac{q_3 - q_2}{k_3 - k_2} = \frac{2800 - 0}{40 - 240} = \frac{2800}{-200} = -14 \text{ km/h}$$

→ The shock front moves upstream at 14km/h.

→ The queue grows against traffic.

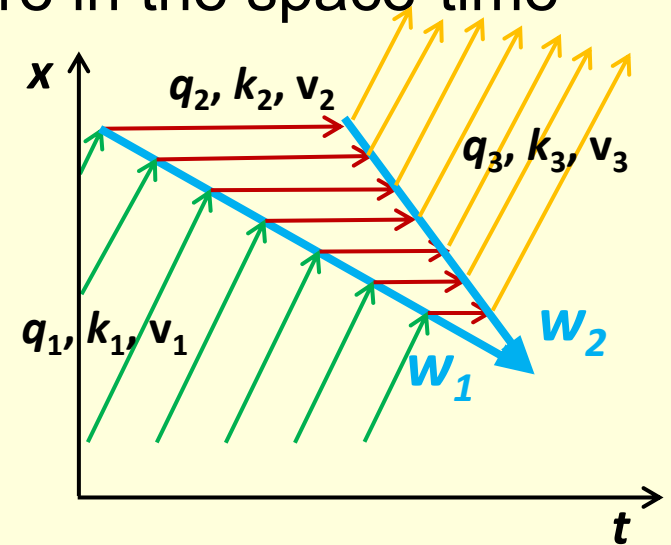


Question

Queue Growth Rate

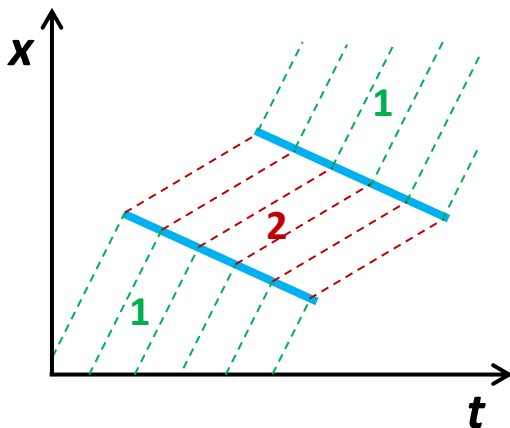
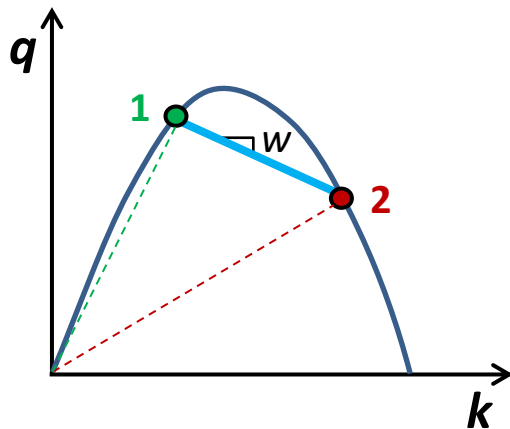
In total, how many shockwaves are there in the space-time diagram?

- A. 2
- B. 3
- C. 4
- D. 5
- E. 6

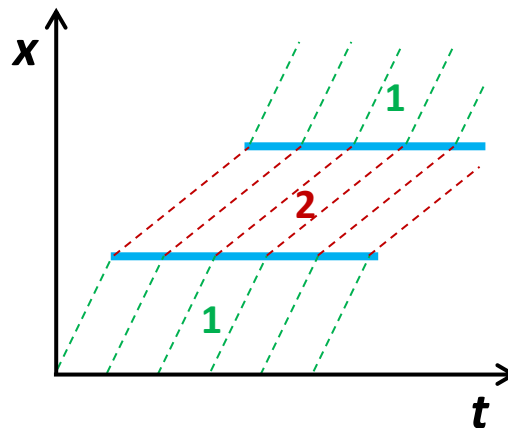
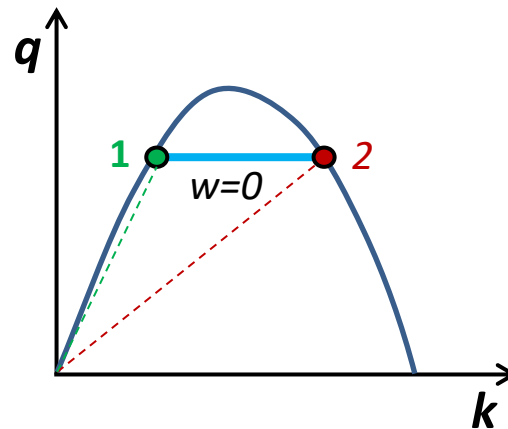


Types of Shockwaves

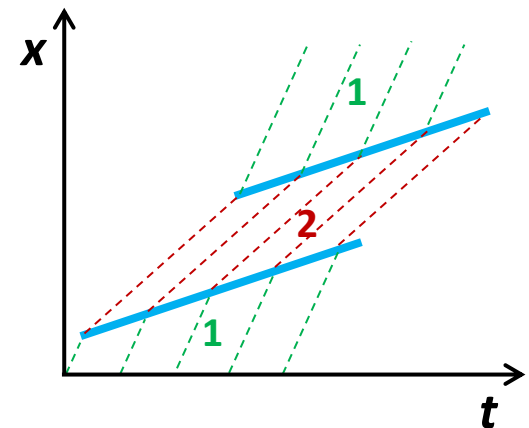
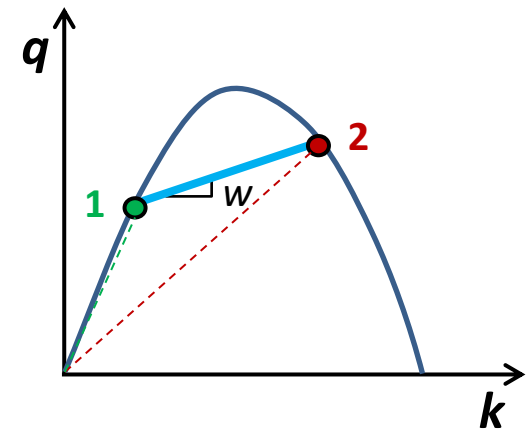
Backward moving shock
 $w < 0$ (negative)



Stationary shockwave
 $w = 0$



Forward moving shock
 $w > 0$ (positive)



Next

This week

- Tutorial – A/D curves